What Cross-Modal Connectome Prediction Does and Does Not Buy

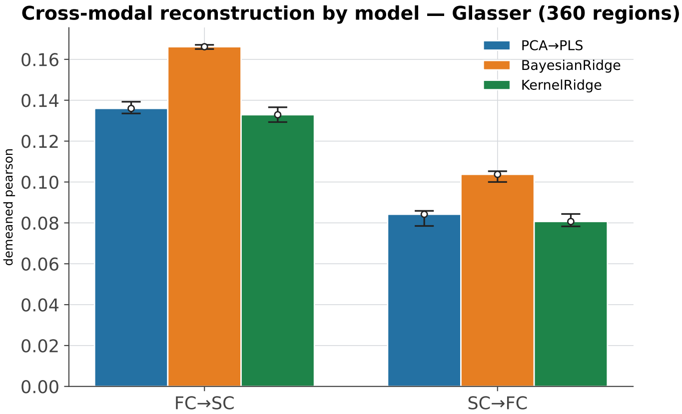
Reconstruction, baselines, and downstream utility (Glasser; 4S456 in backup)

Adel Sahuc

1 · Cross-modal prediction is directional [PCA→PLS]

direction	**demeaned r**
FC→SC	0.136
SC→FC	0.084
ratio	**1.61**×

1 · Cross-modal prediction is directional [PCA→PLS]



Across all three models the FC→SC bars are ∼1.6× taller than SC→FC: FC reconstructs SC's individual deviations far better than the reverse, and the ratio barely changes by model, so it is not an estimator artifact. Whisker = 25th–75th percentile across the 10 seeds; white circle = mean.

2 · A cheap baseline beats cross-modal reconstruction [PCA→PLS]

source	**→ SC**	**→ FC**
cross-modal connectome	0.136	0.084
brain volume (bv)	0.162	0.049
demographics (demo)	0.130	0.111
bv+demo	0.169	0.100
connectome + bv+demo	**0.171**	**0.105**

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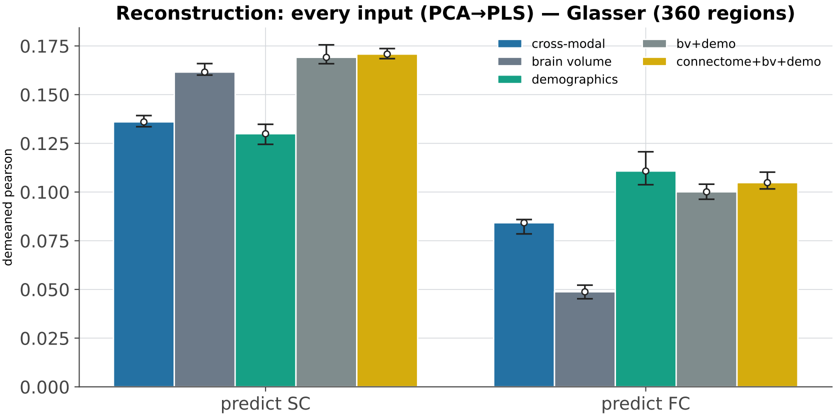
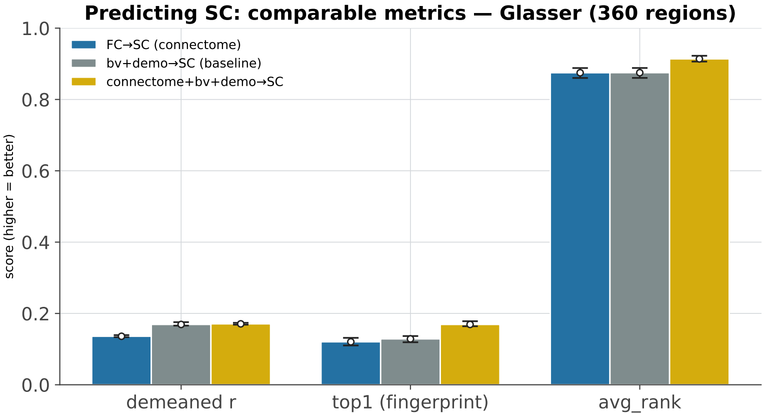


For *both* targets bv+demo matches or beats the cross-modal connectome: a cheap age+sex+volume vector reconstructs the missing connectome as well as the other modality does. Adding the connectome on top of bv+demo (gold) gives only a marginal extra, so the connectome adds little beyond subject information. Note the dissociation: bv (volume) wins for SC, demo wins for FC.

3a · Predicting SC: every metric [PCA→PLS]

input	demeaned r	raw r	top1	avg rank	mse	r^2
FC→SC	0.136	0.913	0.120	0.875	0.006	-0.021
bv→SC	0.162	0.915	0.086	0.864	0.005	0.014
demo→SC	0.130	0.915	0.023	0.754	0.005	0.001
bv+demo→SC	0.169	0.914	0.129	0.875	0.005	0.008
FC+bv+demo→SC	**0.171**	**0.913**	**0.169**	**0.914**	**0.005**	**-0.009**

3a · Predicting SC: comparable metrics [PCA→PLS]

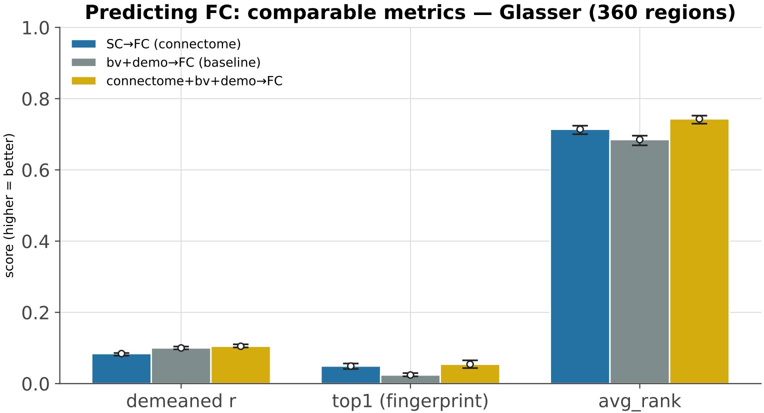


Raw r is ∼0.9 for every input (dominated by the shared group-mean connectome, so uninformative); demeaned r is the honest individual-deviation metric, where bv+demo leads. Whisker = 25th–75th percentile over the 10 seeds.

3b · Predicting FC: every metric [PCA→PLS]

input	demeaned r	raw r	top1	avg rank	mse	r^2
SC→FC	0.084	0.828	0.049	0.714	0.014	-0.056
bv→FC	0.049	0.833	0.012	0.623	0.014	-0.022
demo→FC	0.111	0.837	0.013	0.671	0.014	-0.002
bv+demo→FC	0.100	0.834	0.024	0.685	0.014	-0.019
SC+bv+demo→FC	**0.105**	**0.829**	**0.054**	**0.743**	**0.014**	**-0.047**

3b · Predicting FC: comparable metrics [PCA→PLS]

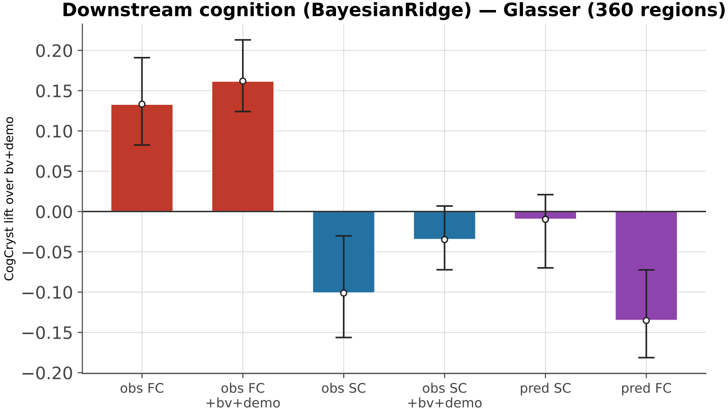


Same picture predicting FC, with one twist: the real connectome (SC) beats bv+demo on the fingerprint metrics (top1, avg rank), i.e. it carries subject-identity information the cheap baseline lacks.

4 · Downstream cognition: only observed FC helps [BayesianRidge]

input	**CogCryst r**	**Total lift**	**Fluid lift**	**Cryst lift**
bv+demo (baseline)	0.354	+0.000	+0.000	+0.000
observed FC	**0.487**	**+0.088**	**+0.044**	**+0.133**
observed SC	0.253	-0.111	-0.120	-0.101
observed FC + SC	0.490	+0.090	+0.048	+0.136
observed FC + bv+demo	**0.516**	**+0.115**	**+0.073**	**+0.162**
observed SC + bv+demo	0.319	-0.050	-0.066	-0.035
predicted SC	0.344	-0.018	-0.053	-0.010
predicted FC	0.219	-0.113	-0.097	-0.135
predicted SC + bv+demo	0.415	+0.040	+0.022	+0.061
predicted FC + bv+demo	0.314	-0.038	-0.038	-0.040

4 · Downstream cognition: only observed FC helps [BayesianRidge]



Only observed FC and observed FC+bv+demo rise above zero (beat the baseline). Observed SC and both predicted connectomes sit at or below zero; predicted FC is actively harmful. Wide whiskers (25–75th pct over seeds) show even the FC lift is variable.

5 · Downstream family: the predicted connectome carries it [PCA→PLS]

predictor	**MZ**	**DZ**	**sibling**
observed SC	0.999	0.938	0.863
observed FC	0.990	0.886	0.823
predicted SC (raw)	0.974	0.798	0.680
predicted SC (demog removed)	**0.989**	**0.823**	**0.810**
predicted FC (raw)	0.910	0.756	0.710
predicted FC (demog removed)	0.918	0.741	0.743
combined predicted SC	**0.897**	**0.701**	**0.505**
bv+demo baseline	0.951	0.800	0.563

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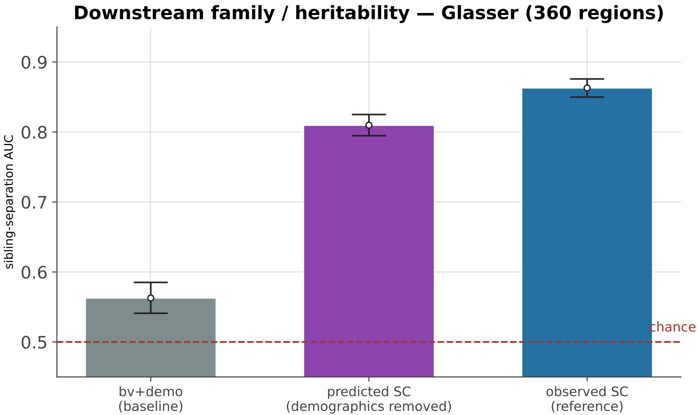

Predicted SC separates siblings far above the bv+demo baseline and close to observed SC: the predicted connectome keeps heritable family structure. Whisker = bootstrap 95% CI (family is a single pipeline with no seed split).

6 · Are the results model-dependent?

Headlines replicate across models; the only divergences are PLS's low oracle and KernelRidge's unstable scalar rows.

reconstruction	PCA→PLS	BayesianRidge	KernelRidge
FC→SC	0.136	0.166	0.133
SC→FC	0.084	0.104	0.081
asymmetry ratio	**1.61**×	**1.60**×	**1.65**×
SC→SC oracle	0.376	0.648	0.648

CogCryst lift	PCA→PLS	BayesianRidge	KernelRidge
obs FC	+0.095	+0.133	+0.181
obs SC	-0.009	-0.101	+0.088
pred FC	-0.097	-0.135	+0.019

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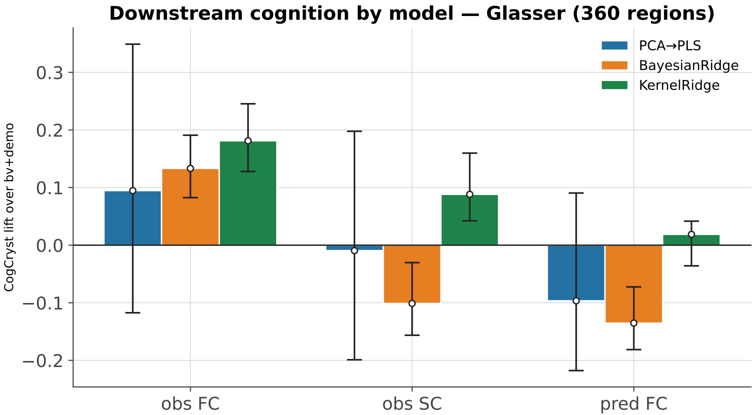


The observed-FC cognition lift is positive under all three models, but the negatives (obs SC, pred FC) flip sign under KernelRidge (green): KernelRidge's scalar regression is numerically unstable, which is why cognition is reported with BayesianRidge.

Takeaways

1. Cross-modal prediction is **real and directional**: FC$\rightarrow$SC $>$ SC$\rightarrow$FC (1.61$\times$).
2. But a **cheap bv+demo baseline beats it** at reconstruction — only the *fingerprint/identity* signal is uniquely connectomic.
3. Downstream, signal splits cleanly: **observed FC** carries (crystallized) cognition; **predicted SC** carries family/heritable signal.

Thank you

Questions?

Backup: deeper dives (Q1, Q2), then 4S456 replication.

Q1 · Can a predicted connectome beat the real one? [BayesianRidge]

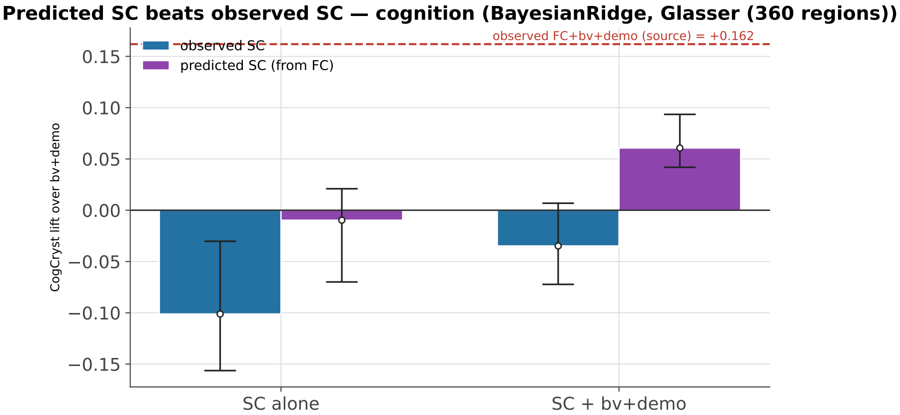
Predicted SC beats *observed* SC for cognition — because predicted SC is a projection of FC (the strong modality). It still loses to FC itself.

input	**Glasser**	**4S456**
observed SC	-0.101	-0.095
predicted SC	-0.010	-0.012
observed SC + bv+demo	-0.035	-0.042
predicted SC + bv+demo	**+0.061**	**+0.056**
(source) observed FC + bv+demo	*+0.162*	*+0.146*

. . . but the flip is estimator-specific:

CogCryst lift (Glasser)	**PCA→PLS**	**BayesianRidge**	**KernelRidge**
predicted SC + bv+demo	+0.019	+0.061	+0.017
observed SC + bv+demo	+0.050	-0.035	+0.128
predicted > observed?	no	**yes**	no

Q1 · Predicted SC beats observed SC — but never the source [BayesianRidge]



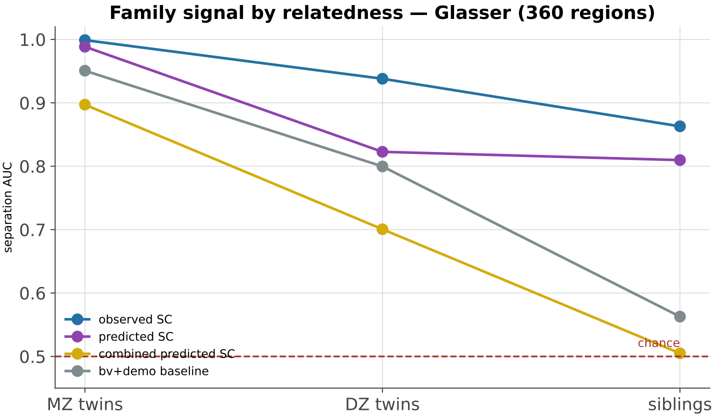
Predicted SC (purple) beats observed SC (blue) because predicted SC is a projection of FC, the modality that carries cognition. It never reaches observed FC itself (dashed line): you would always do better using FC directly.

Q2 · Family signal across relatedness (MZ / DZ / sibling) [PCA→PLS]

A genetic dose-response; the reconstruction-optimized *combined* predictor collapses to chance for siblings.

predictor	**MZ twins**	**DZ twins**	**siblings**
observed SC	0.999	0.938	0.863
predicted SC	0.989	0.823	0.810
predicted FC	0.918	0.741	0.743
combined predicted SC	0.897	0.701	0.505
bv+demo baseline	0.951	0.800	0.563

Q2 · Heritability gradient & the objective tradeoff [PCA→PLS]



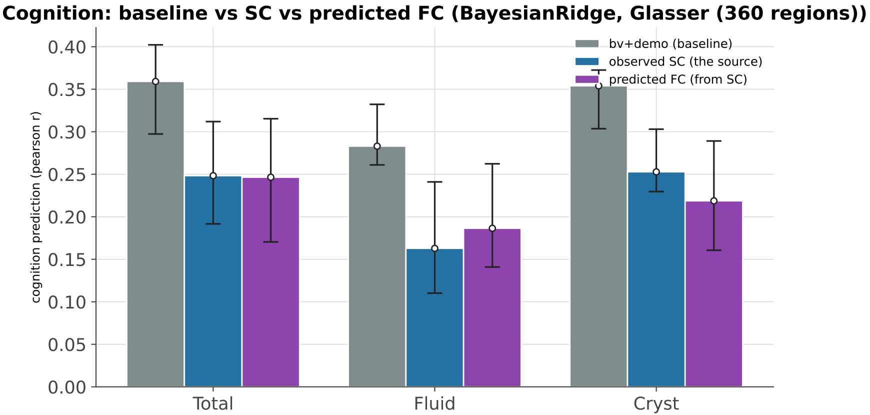
Every predictor weakens from MZ twins to DZ twins to siblings (a genetic dose-response). The combined predictor (gold) falls to the chance line at siblings: optimizing for reconstruction discards the fine-grained family signal.

Q3 · Can a prediction beat the connectome it came from? [BayesianRidge]

Predicted FC beats its source SC on the noisy targets (a denoising effect) — but both stay below the baseline; predicted SC never beats its source FC.

input (CogXxx pearson r)	**CogTotal**	**CogFluid**	**CogCryst**
bv+demo (baseline)	0.359	0.283	0.354
observed FC	0.447	0.327	0.487
→ predicted SC	0.341	0.230	0.344
observed FC + bv+demo	0.474	0.356	0.516
→ predicted SC + bv+demo	0.399	0.305	0.415
observed SC	0.248	0.163	0.253
→ predicted FC	0.247	**0.186**	0.219
observed SC + bv+demo	0.309	0.217	0.319
→ predicted FC + bv+demo	**0.321**	**0.245**	0.314

Q3 · Prediction beats source only as ‘less harmful’ [BayesianRidge]



Every connectome row sits below the bv+demo baseline (grey, tallest). ‘Beating the source’ just means less far below: predicted FC edges observed SC on CogFluid, and predicted FC+bv+demo beats observed SC+bv+demo on CogTotal and CogFluid (see table) — a mild denoising of the weak modality. Predicted SC never beats observed FC, the strong modality.

1 · Cross-modal prediction is directional [PCA→PLS] (4S456 backup)

direction	**demeaned r**
FC→SC	0.146
SC→FC	0.087
ratio	**1.68**×

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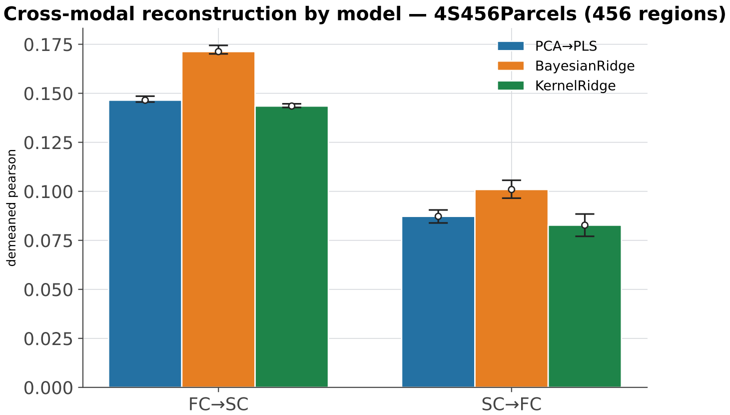

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2 · A cheap baseline beats cross-modal reconstruction [PCA→PLS] (4S456 backup)

source	**→ SC**	**→ FC**
cross-modal connectome	0.146	0.087
brain volume (bv)	0.185	0.045
demographics (demo)	0.133	0.105
bv+demo	0.191	0.094
connectome + bv+demo	**0.184**	**0.106**

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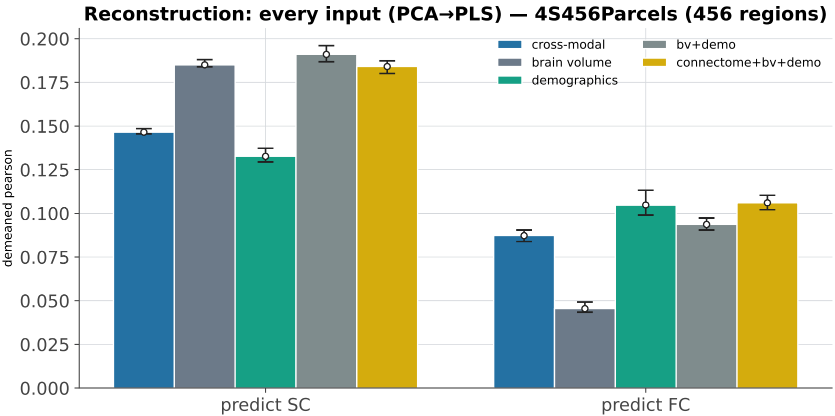

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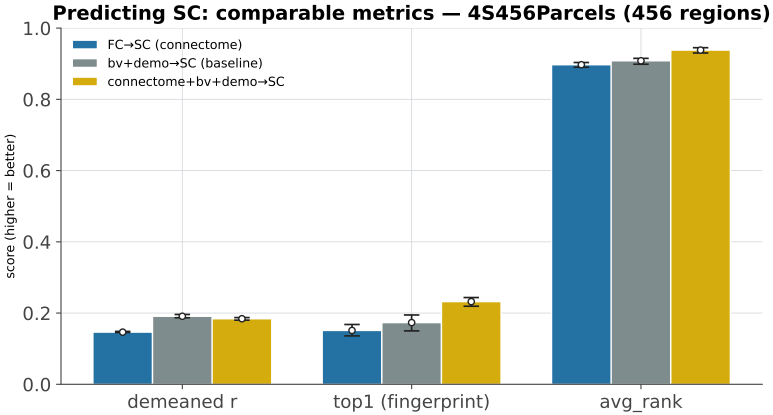
3a · Predicting SC: every metric [PCA→PLS] (4S456 backup)

input	demeaned r	raw r	top1	avg rank	mse	r^2
FC→SC	0.146	0.911	0.151	0.897	0.004	-0.017
bv→SC	0.185	0.913	0.129	0.903	0.004	0.015
demo→SC	0.133	0.912	0.023	0.764	0.004	-0.001
bv+demo→SC	0.191	0.913	0.173	0.908	0.004	0.010
FC+bv+demo→SC	**0.184**	**0.912**	**0.232**	**0.938**	**0.004**	**-0.004**

3a · Predicting SC: comparable metrics [PCA→PLS] (4S456 backup)



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3b · Predicting FC: every metric [PCA→PLS] (4S456 backup)

input	demeaned r	raw r	top1	avg rank	mse	r^2
SC→FC	0.087	0.813	0.049	0.730	0.013	-0.047
bv→FC	0.045	0.817	0.010	0.625	0.012	-0.022
demo→FC	0.105	0.821	0.013	0.677	0.012	-0.004
bv+demo→FC	0.094	0.818	0.024	0.687	0.012	-0.020
SC+bv+demo→FC	**0.106**	**0.815**	**0.055**	**0.758**	**0.013**	**-0.040**

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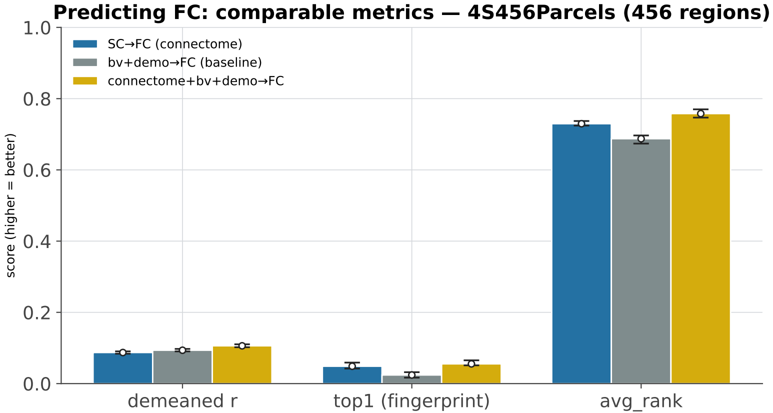

Same picture predicting FC, with one twist: the real connectome (SC) beats bv+demo on the fingerprint metrics (top1, avg rank), i.e. it carries subject-identity information the cheap baseline lacks.

4 · Downstream cognition: only observed FC helps [BayesianRidge] (4S456 backup)

input	CogCryst r	Total lift	Fluid lift	Cryst lift
bv+demo (baseline)	0.354	+0.000	+0.000	+0.000
observed FC	**0.476**	**+0.086**	**+0.037**	**+0.122**
observed SC	0.259	-0.103	-0.118	-0.095
observed FC + SC	0.475	+0.087	+0.043	+0.121
observed FC + bv+demo	**0.500**	**+0.109**	**+0.062**	**+0.146**
observed SC + bv+demo	0.312	-0.052	-0.077	-0.042
predicted SC	0.342	-0.005	-0.010	-0.012
predicted FC	0.245	-0.088	-0.072	-0.109
predicted SC + bv+demo	0.410	+0.048	+0.042	+0.056
predicted FC + bv+demo	0.326	-0.025	-0.024	-0.028

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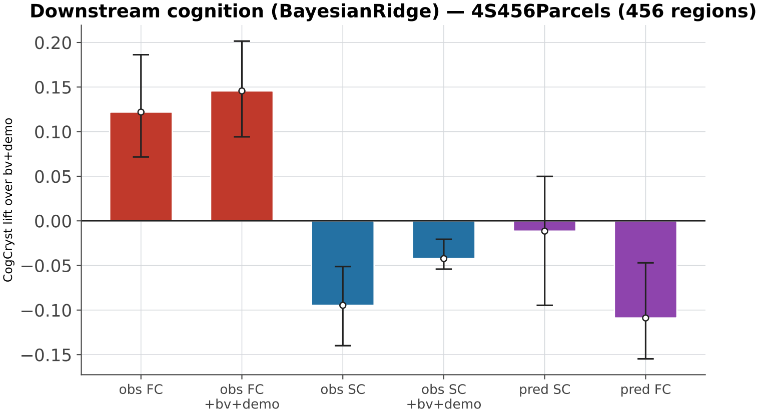

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5 · Downstream family: the predicted connectome carries it [PCA→PLS] (4S456 backup)

predictor	MZ	DZ	sibling
observed SC	0.999	0.958	0.884
observed FC	0.991	0.886	0.821
predicted SC (raw)	0.981	0.793	0.670
predicted SC (demog removed)	**0.994**	**0.847**	**0.816**
predicted FC (raw)	0.967	0.807	0.726
predicted FC (demog removed)	0.966	0.797	0.769
combined predicted SC	**0.908**	**0.691**	**0.501**
bv+demo baseline	0.949	0.791	0.565

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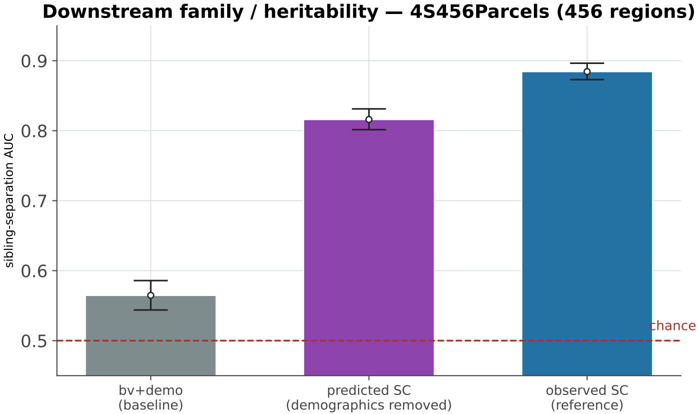

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FC→SC	0.146	0.171	0.143
SC→FC	0.087	0.101	0.083
asymmetry ratio	**1.68**×	**1.70**×	**1.73**×
SC→SC oracle	0.386	0.622	0.621

CogCryst lift	**PCA→PLS**	**BayesianRidge**	**KernelRidge**
obs FC	+0.137	+0.122	+0.176
obs SC	-0.047	-0.095	+0.094
pred FC	-0.101	-0.109	-0.014

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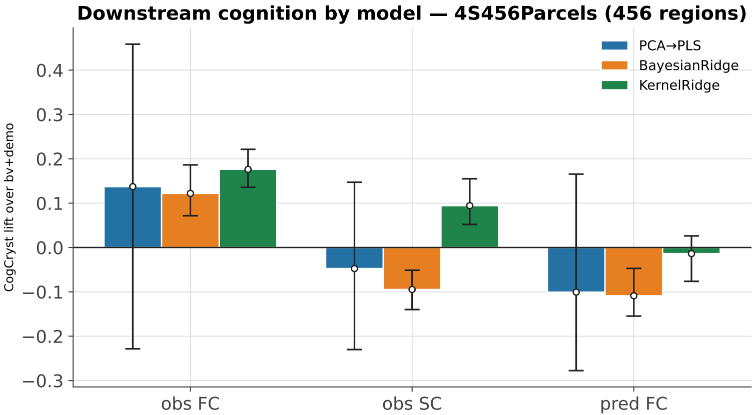

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